

TrES-3b

R-0449

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180707 · created 2026-07-19T16:38:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458306.9284 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.055
Transit depth (Rp/Rs)^2	3.23 +/- 1.96 [%]
Orbital Inclination (inc)	80.12 +/- 2.77
Airmass coefficient 1 (a1)	0.977 +/- 0.023
Airmass coefficient 2 (a2)	0.017 +/- 0.017
Scatter in the residuals of the lightcurve fit is	2.34 %
Best Comparison Star	#1 - [379, 431]
Optimal Method	PSF photometry
Transit Duration (day)	0.033 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.18 ± 0.055 vs 0.16309 (deviation 0.31σ)
Duration fit vs published	0.033 d vs 0.0567 d (deviation 0.85σ)
Mid-time O-C	-1.4 min
Depth SNR	3.3
Residual scatter	2.34 %

Light curve

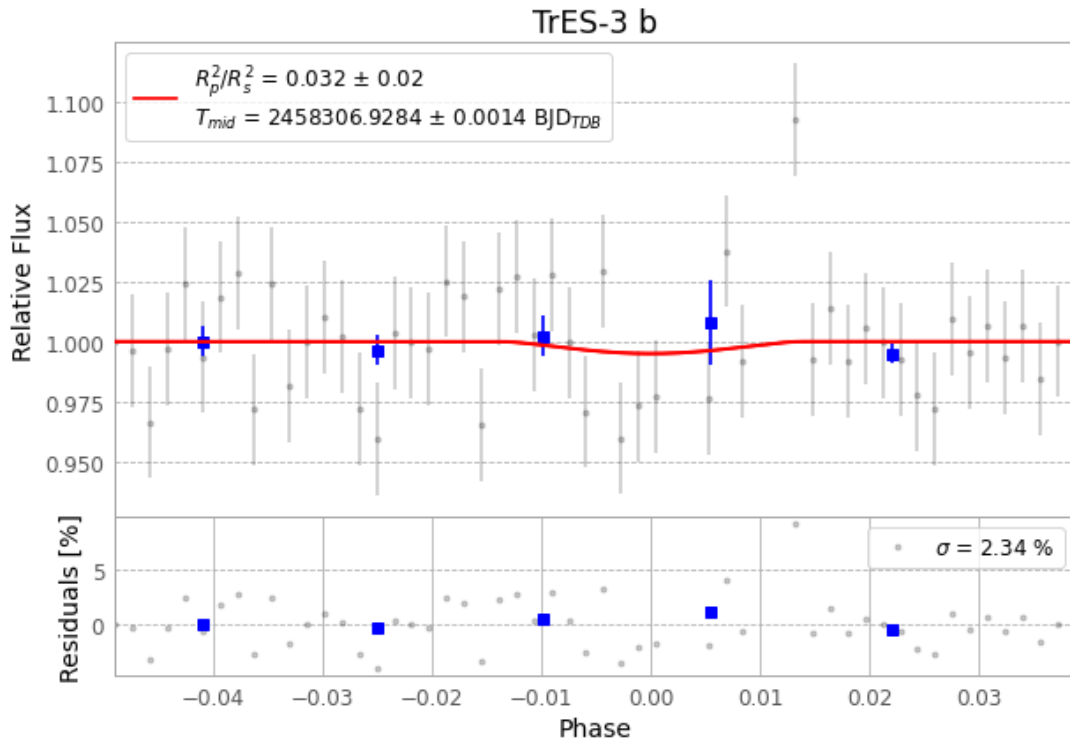


Figure 1 — FinalLightCurve_TrES-3 b_2018-07-07.png (SHA-256 e79e134318de66da...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [271, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 97950ae35ebd29ed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

56 FITS frames (37 MB), TRES-3180707083912.FITS → TRES-3180707112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180707.log (fba0b4b70885...), FinalLightCurve_TrES-3 b_2018-07-07.png (e79e134318de...), FinalLightCurve_TrES-3 b_2018-07-07.csv (4ce5654f74d8...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 16:38Z.